R Data Structure and Tidy Data

PSYC 259: Principles of Data Science

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Admin for Today

- Workflow critique
- Resources on course site

Today's Outline

- Part 1: More advanced data types
  - Strings and logicals
  - Dates
  - Factors
  - Vectors and lists
- Part 2: Tidying data
  - Pivoting from/to wide/long format
  - Binding and joining data

Follow along from the Github repo

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Part 1: More Advanced Data Types

Common Data Types in R

- Numeric
  - integer: 1, 2, 3
  - double: 1.12124
- Character (aka string)
  - "hello"
- Logical
  - T/F (TRUE/FALSE)
- Vectors and lists
  - Two different ways to store and manipulate data
- Datetime
  - "2026-10-18"
- Factor
  - eye_colors <- c("Blue", "Green", "Brown")
  - factor(eye_colors)

What Packages Do We Need?

- Default tidyverse
  - `ggplot2`: Graphing
  - `tibble`: Using tibble-style dataframes
  - `tidyr`: Shaping tibbles and columns
  - `purrr`: Automating/iterating (instead of loops)
  - `dplyr`: Selecting, filtering, and transforming dataframes
  - `stringr`: Manipulating strings
  - `forcats`: Manipulating factors
  - `lubridate`: Manipulating dates

Character Data (Strings)

- At minimum, working with characters is needed to deal with file paths
- String functions let you clean up messy text input
- Understanding how to construct and format strings can help you avoid hard-coding things like graph labels and output tables
- Learning how to wrangle strings and exploit patterns helps write more efficient code (e.g., selecting and mutating columns with `contains()`, `starts_with()`, etc.)
- The `stringr` package has a suite of helper functions for common string operations

stringr Cheatsheet

View on RStudio GitHub

Jonah's first words

```
ds$word
```

```
 [1] "book"              "dad"               "cat"
 [4] "baby"              "socks/shoes"       "banana"
 [7] "cheese"            "shoes"             "monkey"
[10] "hello"             "bear"              "milk/drink"
[13] "cleanup"           "grandpa"           "mouth"
[16] "coffee"            "agua"              "octopus"
[19] "(wrist) watch"     "shower"            "hyena"
[22] "heart"             "singing"           "top"
[25] "ball"              "dog (animal)"      "llama"
[28] "turkey"            "berry"             "apple"
[31] "oh no"             "go"                "car"
[34] "that"              "more"              "peacock"
[37] "cup"               "hide"              "too"
[40] "blanket"           "peanut butter"     "running"
[43] "empty"             "that one"          "sandal"
[46] "bye bye"           "beep"              "hi"
[49] "box"               "night-night"       "jacket"
[52] "dinosaur"          "wow"               "snack/food"
[55] "wash"              "stroller"          "shark"
[58] "grandma"           "face"              "lemon"
[61] "potty"             "the end"           "over there"
[64] "make"              "diaper"            "bubble"
[67] "fish"              "tea"               "peekaboo"
```

Length, Detecting Strings, & Counting Strings

```
ds %>% mutate(word_length = str_length(word),
              has_b = str_detect(word, "b"),
              num_b = str_count(word, "b"))
```

```
# A tibble: 440 × 5
   age   word        word_length has_b num_b
   <chr> <chr>             <int> <lgl> <int>
 1 12    book                  4 TRUE      1
 2 12.5  dad                   3 FALSE     0
 3 13.5  cat                   3 FALSE     0
 4 14    baby                  4 TRUE      2
 5 14.5  socks/shoes          11 FALSE     0
 6 15    banana                6 TRUE      1
 7 15.5  cheese                6 FALSE     0
 8 16    shoes                 5 FALSE     0
 9 16.5  monkey                6 FALSE     0
10 17    hello                 5 FALSE     0
# ℹ 430 more rows
```

Regular Expressions (regex)

- A pattern language for matching text, used in `stringr` functions like `str_detect()`, `str_extract()`, `str_replace_all()`
- In R strings, a backslash must be doubled: `"\\d"`, `"\\."`

Characters and classes

- `.` any character
- `\\d` digit, `\\w` word character, `\\s` space
- `[aeiou]` any one of these
- `[^aeiou]` anything but these
- `[:alpha:]`, `[:digit:]`, `[:punct:]` anything in this class

Anchors

- `^` start of string
- `$` end of string
- `\\b` word boundary

Regular Expressions (regex)

Quantifiers

- `?` 0 or 1
- `*` 0 or more
- `+` 1 or more
- `{3}` exactly 3, `{2,4}` 2 to 4

Alternatives and groups

- `a|b` a or b
- `(ab)+` group, then repeat
- `\\.` a literal period (escape special characters)

RStudio regex cheatsheet

Matching Patterns

```
ds %>% mutate(contains_punct = str_detect(word, "[:punct:]"))
```

```
# A tibble: 440 × 3
   age   word        contains_punct
   <chr> <chr>       <lgl>
 1 12    book        FALSE
 2 12.5  dad         FALSE
 3 13.5  cat         FALSE
 4 14    baby        FALSE
 5 14.5  socks/shoes TRUE
 6 15    banana      FALSE
 7 15.5  cheese      FALSE
 8 16    shoes       FALSE
 9 16.5  monkey      FALSE
10 17    hello       FALSE
# ℹ 430 more rows
```

Matching Patterns

```
ds %>% mutate(word_cleaned = str_remove_all(word, "[:punct:]"))
```

```
# A tibble: 440 × 3
   age   word        word_cleaned
   <chr> <chr>       <chr>
 1 12    book        book
 2 12.5  dad         dad
 3 13.5  cat         cat
 4 14    baby        baby
 5 14.5  socks/shoes socksshoes
 6 15    banana      banana
 7 15.5  cheese      cheese
 8 16    shoes       shoes
 9 16.5  monkey      monkey
10 17    hello       hello
# ℹ 430 more rows
```

Matching Patterns

```
ds %>% mutate(word_cleaned = str_replace_all(word, "[:punct:]"," "))
```

```
# A tibble: 440 × 3
   age   word        word_cleaned
   <chr> <chr>       <chr>
 1 12    book        book
 2 12.5  dad         dad
 3 13.5  cat         cat
 4 14    baby        baby
 5 14.5  socks/shoes socks shoes
 6 15    banana      banana
 7 15.5  cheese      cheese
 8 16    shoes       shoes
 9 16.5  monkey      monkey
10 17    hello       hello
# ℹ 430 more rows
```

Matching Patterns

```
ds %>% mutate(vowel_count = str_count(word, "[aeiou]"))
```

```
# A tibble: 440 × 3
   age   word        vowel_count
   <chr> <chr>             <int>
 1 12    book                  2
 2 12.5  dad                   1
 3 13.5  cat                   1
 4 14    baby                  1
 5 14.5  socks/shoes           3
 6 15    banana                3
 7 15.5  cheese                3
 8 16    shoes                 2
 9 16.5  monkey                2
10 17    hello                 2
# ℹ 430 more rows
```

```r
fnames <- list.files(path="data_raw", full.names = T)
print(fnames)
```

```
 [1] "data_raw/vocab12.5.csv" "data_raw/vocab12.csv"   "data_raw/vocab13.5.csv"
 [4] "data_raw/vocab14.5.csv" "data_raw/vocab14.csv"   "data_raw/vocab15.5.csv"
 [7] "data_raw/vocab15.csv"   "data_raw/vocab16.5.csv" "data_raw/vocab16.csv"
[10] "data_raw/vocab17.5.csv" "data_raw/vocab17.csv"   "data_raw/vocab18.5.csv"
[13] "data_raw/vocab18.csv"   "data_raw/vocab19.5.csv" "data_raw/vocab19.csv"
[16] "data_raw/vocab20.5.csv" "data_raw/vocab20.csv"   "data_raw/vocab21.5.csv"
[19] "data_raw/vocab21.csv"   "data_raw/vocab22.5.csv" "data_raw/vocab22.csv"
[22] "data_raw/vocab23.5.csv" "data_raw/vocab23.csv"   "data_raw/vocab24.csv"
```

Matching Patterns

```
fnames <- list.files(path="data_raw", full.names = T)
str_split(fnames, "/")
```

```
[[1]]
[1] "data_raw"      "vocab12.5.csv"

[[2]]
[1] "data_raw"    "vocab12.csv"

[[3]]
[1] "data_raw"      "vocab13.5.csv"

[[4]]
[1] "data_raw"      "vocab14.5.csv"

[[5]]
[1] "data_raw"    "vocab14.csv"

[[6]]
[1] "data_raw"      "vocab15.5.csv"

[[7]]
[1] "data_raw"    "vocab15.csv"

[[8]]
[1] "data_raw"      "vocab16.5.csv"

[[9]]
[1] "data_raw"    "vocab16.csv"

[[10]]
[1] "data_raw"      "vocab17.5.csv"
```

Matching Patterns

```
fnames <- list.files(path="data_raw", full.names = T)
str_replace_all(fnames, "data_raw/|.csv|vocab","")
```

```
 [1] "12.5" "12"   "13.5" "14.5" "14"   "15.5" "15"   "16.5" "16"   "17.5"
[11] "17"   "18.5" "18"   "19.5" "19"   "20.5" "20"   "21.5" "21"   "22.5"
[21] "22"   "23.5" "23"   "24"
```

Matching Patterns

```
fnames <- list.files(path="data_raw", full.names = T)
str_extract_all(fnames, "\\d{2}.\\d{1}|\\d{2}", simplify = T)
```

```
      [,1]
 [1,] "12.5"
 [2,] "12"
 [3,] "13.5"
 [4,] "14.5"
 [5,] "14"
 [6,] "15.5"
 [7,] "15"
 [8,] "16.5"
 [9,] "16"
[10,] "17.5"
[11,] "17"
[12,] "18.5"
[13,] "18"
[14,] "19.5"
[15,] "19"
[16,] "20.5"
[17,] "20"
[18,] "21.5"
[19,] "21"
[20,] "22.5"
[21,] "22"
[22,] "23.5"
[23,] "23"
[24,] "24"
```

Matching Patterns

```
fnames <- list.files(path="data_raw", full.names = T)
str_extract_all(fnames, "\\.[:alpha:]{1,4}", simplify = T)
```

```
      [,1]
 [1,] ".csv"
 [2,] ".csv"
 [3,] ".csv"
 [4,] ".csv"
 [5,] ".csv"
 [6,] ".csv"
 [7,] ".csv"
 [8,] ".csv"
 [9,] ".csv"
[10,] ".csv"
[11,] ".csv"
[12,] ".csv"
[13,] ".csv"
[14,] ".csv"
[15,] ".csv"
[16,] ".csv"
[17,] ".csv"
[18,] ".csv"
[19,] ".csv"
[20,] ".csv"
[21,] ".csv"
[22,] ".csv"
[23,] ".csv"
[24,] ".csv"
```

Matching Patterns

```
fnames <- list.files(path="data_raw", full.names = T)
str_extract_all(ds$word, boundary("word"), simplify = T)
```

```
      [,1]        [,2]
 [1,] "book"      ""
 [2,] "dad"       ""
 [3,] "cat"       ""
 [4,] "baby"      ""
 [5,] "socks"     "shoes"
 [6,] "banana"    ""
 [7,] "cheese"    ""
 [8,] "shoes"     ""
 [9,] "monkey"    ""
[10,] "hello"     ""
[11,] "bear"      ""
[12,] "milk"      "drink"
[13,] "cleanup"   ""
[14,] "grandpa"   ""
[15,] "mouth"     ""
[16,] "coffee"    ""
[17,] "agua"      ""
[18,] "octopus"   ""
[19,] "wrist"     "watch"
[20,] "shower"    ""
[21,] "hyena"     ""
[22,] "heart"     ""
[23,] "singing"   ""
[24,] "top"       ""
[25,] "ball"      ""
[26,] "dog"       "animal"
[27,] "llama"     ""
[28,] "turkey"    ""
```

Combining Strings

```
age <- seq(from = 12, to = 24, by = .5)
paste0("data_raw/vocab", age, ".csv")
```

```
 [1] "data_raw/vocab12.csv"   "data_raw/vocab12.5.csv" "data_raw/vocab13.csv"
 [4] "data_raw/vocab13.5.csv" "data_raw/vocab14.csv"   "data_raw/vocab14.5.csv"
 [7] "data_raw/vocab15.csv"   "data_raw/vocab15.5.csv" "data_raw/vocab16.csv"
[10] "data_raw/vocab16.5.csv" "data_raw/vocab17.csv"   "data_raw/vocab17.5.csv"
[13] "data_raw/vocab18.csv"   "data_raw/vocab18.5.csv" "data_raw/vocab19.csv"
[16] "data_raw/vocab19.5.csv" "data_raw/vocab20.csv"   "data_raw/vocab20.5.csv"
[19] "data_raw/vocab21.csv"   "data_raw/vocab21.5.csv" "data_raw/vocab22.csv"
[22] "data_raw/vocab22.5.csv" "data_raw/vocab23.csv"   "data_raw/vocab23.5.csv"
[25] "data_raw/vocab24.csv"
```

Combining Strings

```
age <- seq(from = 12, to = 24, by = .5)
str_glue("data_raw/vocab{age}.csv")
```

```
data_raw/vocab12.csv
data_raw/vocab12.5.csv
data_raw/vocab13.csv
data_raw/vocab13.5.csv
data_raw/vocab14.csv
data_raw/vocab14.5.csv
data_raw/vocab15.csv
data_raw/vocab15.5.csv
data_raw/vocab16.csv
data_raw/vocab16.5.csv
data_raw/vocab17.csv
data_raw/vocab17.5.csv
data_raw/vocab18.csv
data_raw/vocab18.5.csv
data_raw/vocab19.csv
data_raw/vocab19.5.csv
data_raw/vocab20.csv
data_raw/vocab20.5.csv
data_raw/vocab21.csv
data_raw/vocab21.5.csv
data_raw/vocab22.csv
data_raw/vocab22.5.csv
data_raw/vocab23.csv
data_raw/vocab23.5.csv
data_raw/vocab24.csv
```

Logical (T/F)

- Logical data are usually created by comparisons or tests in R rather than data we collected
    - Created by operators like `>`,`<`,`==`,`!=`,`>=`, `%in%`, `is.na()`, `str_detect()`, etc.
    - Also inside mutate(), filter(), etc.
- FALSE = 0, TRUE = 1
    - sum() counts matches and mean() gives a proportion of TRUEs

Logicals from Text

```
contains_b <- str_detect(ds$word, "b")  # TRUE if the word contains b
```

Logicals from Text

```
contains_b <- str_detect(ds$word, "b")  # TRUE if the word contains b
contains_b
```

Logicals from Text

```
contains_b <- str_detect(ds$word, "b")  # TRUE if the word contains b
contains_b
typeof(contains_b)
```

Logicals from Text

```
contains_b <- str_detect(ds$word, "b")  # TRUE if the word contains b
contains_b
typeof(contains_b)
sum(contains_b)
```

Logicals from Text

```
contains_b <- str_detect(ds$word, "b")  # TRUE if the word contains b
contains_b
typeof(contains_b)
sum(contains_b)
mean(contains_b)
```

Logicals from Text

```
contains_b <- str_detect(ds$word, "b")  # TRUE if the word contains b
contains_b
typeof(contains_b)
sum(contains_b)
mean(contains_b)
!contains_b
```

Logicals from Text

```
contains_b <- str_detect(ds$word, "b")  # TRUE if the word contains b
contains_b
typeof(contains_b)
sum(contains_b)
mean(contains_b)
!contains_b
ds$word[contains_b]
```

Logicals from Text

```
contains_b <- str_detect(ds$word, "b")  # TRUE if the word contains b
contains_b
typeof(contains_b)
sum(contains_b)
mean(contains_b)
!contains_b
ds$word[contains_b]
ds %>% mutate(contains_b = contains_b)
```

Logicals from Text

```
contains_b <- str_detect(ds$word, "b")  # TRUE if the word contains b
contains_b
typeof(contains_b)
sum(contains_b)
mean(contains_b)
!contains_b
ds$word[contains_b]
ds %>% mutate(contains_b = contains_b)
ds %>% filter(contains_b)
```

Logicals from Text

```
contains_b <- str_detect(ds$word, "b")  # TRUE if the word contains b
contains_b
typeof(contains_b)
sum(contains_b)
mean(contains_b)
!contains_b
ds$word[contains_b]
ds %>% mutate(contains_b = contains_b)
ds %>% filter(contains_b)
ds %>% mutate(b_and_vowels = contains_b & str_detect(word, "[aeiou]"))
```

Logicals from Text

```
contains_b <- str_detect(ds$word, "b")  # TRUE if the word contains b
contains_b
typeof(contains_b)
sum(contains_b)
mean(contains_b)
!contains_b
ds$word[contains_b]
ds %>% mutate(contains_b = contains_b)
ds %>% filter(contains_b)
ds %>% mutate(b_and_vowels = contains_b & str_detect(word, "[aeiou]"))
ds <- ds %>% mutate(b_or_startswithd = contains_b | str_starts(word, "d"))
```

Vectors and Lists: Collections of Values

- Vectors are type-homogeneous
  - Numeric vectors, `c(1,2,3,4,5)`, `1:5`
  - Character vectors, `c("Hello", "Goodbye")`
  - Logical vectors, `c(T, T, T, F, T, F)`
  - Vector items can be named `c(x = 1, y = 2)`
- Lists are type-heterogeneous
  - Mix types, `list("1", 1, TRUE, 1:5)`
  - Tibbles are lists of vectors/lists of the same length

Vectors

```
v <- c(1, 2, 3, 4, 5)
```

Vectors

```
v <- c(1, 2, 3, 4, 5)
v
```

```
[1] 1 2 3 4 5
```

Vectors

```
v <- c(1, 2, 3, 4, 5)
v
typeof(v)
```

```
[1] 1 2 3 4 5

[1] "double"
```

Vectors

```
v <- c(1, 2, 3, 4, 5)
v
typeof(v)
length(v)
```

```
[1] 1 2 3 4 5

[1] "double"

[1] 5
```

Vectors

```r
v <- c(1, 2, 3, 4, 5)
v
typeof(v)
length(v)
v[1]  # Use brackets to access a vector element
```

```
[1] 1 2 3 4 5
```

```
[1] "double"
```

```
[1] 5
```

```
[1] 1
```

Vectors

```
v <- c(1, 2, 3, 4, 5)
v
typeof(v)
length(v)
v[1]  # Use brackets to access a vector element
v[3]
```

```
[1] 1 2 3 4 5

[1] "double"

[1] 5

[1] 1

[1] 3
```

Vectors

```
v <- c(1, 2, 3, 4, 5)
v
typeof(v)
length(v)
v[1]  # Use brackets to access a vector element
v[3]
v <- c(v, 6, 7, 8)  # use c to add items
```

```
[1] 1 2 3 4 5

[1] "double"

[1] 5

[1] 1

[1] 3
```

Vectors

```
v <- c(1, 2, 3, 4, 5)
v
typeof(v)
length(v)
v[1]  # Use brackets to access a vector element
v[3]
v <- c(v, 6, 7, 8)  # use c to add items
v <- c(1, 2, 3, "4", "5", "6")
```

```
[1] 1 2 3 4 5

[1] "double"

[1] 5

[1] 1

[1] 3
```

Vectors

```r
v <- c(1, 2, 3, 4, 5)
v
typeof(v)
length(v)
v[1]  # Use brackets to access a vector element
v[3]
v <- c(v, 6, 7, 8)  # use c to add items
v <- c(1, 2, 3, "4", "5", "6")
v  # You cannot mix types in a vector
```

```
[1] 1 2 3 4 5

[1] "double"

[1] 5

[1] 1

[1] 3

[1] "1" "2" "3" "4" "5" "6"
```

Vectors

```r
v <- c(1, 2, 3, 4, 5)
v
typeof(v)
length(v)
v[1]  # Use brackets to access a vector element
v[3]
v <- c(v, 6, 7, 8)  # use c to add items
v <- c(1, 2, 3, "4", "5", "6")
v  # You cannot mix types in a vector
v <- c(x = 1, y = 2, z = 3)  # Named vector
```

```
[1] 1 2 3 4 5

[1] "double"

[1] 5

[1] 1

[1] 3

[1] "1" "2" "3" "4" "5" "6"
```

Vectors

```r
v <- c(1, 2, 3, 4, 5)
v
typeof(v)
length(v)
v[1]  # Use brackets to access a vector element
v[3]
v <- c(v, 6, 7, 8)  # use c to add items
v <- c(1, 2, 3, "4", "5", "6")
v  # You cannot mix types in a vector
v <- c(x = 1, y = 2, z = 3)  # Named vector
v
```

```
[1] 1 2 3 4 5

[1] "double"

[1] 5

[1] 1

[1] 3

[1] "1" "2" "3" "4" "5" "6"

x y z
1 2 3
```

Vectors

```r
v <- c(1, 2, 3, 4, 5)
v
typeof(v)
length(v)
v[1]  # Use brackets to access a vector element
v[3]
v <- c(v, 6, 7, 8)  # use c to add items
v <- c(1, 2, 3, "4", "5", "6")
v  # You cannot mix types in a vector
v <- c(x = 1, y = 2, z = 3)  # Named vector
v
v['y']
```

```
[1] 1 2 3 4 5

[1] "double"

[1] 5

[1] 1

[1] 3

[1] "1" "2" "3" "4" "5" "6"

x y z
1 2 3

y
2
```

Vectors

```
v <- c(1, 2, 3, 4, 5)
v
typeof(v)
length(v)
v[1]  # Use brackets to access a vector element
v[3]
v <- c(v, 6, 7, 8)  # use c to add items
v <- c(1, 2, 3, "4", "5", "6")
v  # You cannot mix types in a vector
v <- c(x = 1, y = 2, z = 3)  # Named vector
v
v['y']

files <- list.files(path = "data_raw")
```

```
[1] 1 2 3 4 5

[1] "double"

[1] 5

[1] 1

[1] 3

[1] "1" "2" "3" "4" "5" "6"

x y z
1 2 3

y
2
```

Vectors

```r
v <- c(1, 2, 3, 4, 5)
v
typeof(v)
length(v)
v[1]  # Use brackets to access a vector element
v[3]
v <- c(v, 6, 7, 8)  # use c to add items
v <- c(1, 2, 3, "4", "5", "6")
v  # You cannot mix types in a vector
v <- c(x = 1, y = 2, z = 3)  # Named vector
v
v['y']

files <- list.files(path = "data_raw")
files[1]  # Character vector
```

```
[1] 1 2 3 4 5

[1] "double"

[1] 5

[1] 1

[1] 3

[1] "1" "2" "3" "4" "5" "6"

x y z
1 2 3

y
2

[1] "vocab12.5.csv"
```

Vectors

```r
v <- c(1, 2, 3, 4, 5)
v
typeof(v)
length(v)
v[1]  # Use brackets to access a vector element
v[3]
v <- c(v, 6, 7, 8)  # use c to add items
v <- c(1, 2, 3, "4", "5", "6")
v  # You cannot mix types in a vector
v <- c(x = 1, y = 2, z = 3)  # Named vector
v
v['y']

files <- list.files(path = "data_raw")
files[1]  # Character vector
files[2]
```

```
[1] 1 2 3 4 5

[1] "double"

[1] 5

[1] 1

[1] 3

[1] "1" "2" "3" "4" "5" "6"

x y z
1 2 3

y
2

[1] "vocab12.5.csv"

[1] "vocab12.csv"
```

Vectors

```r
v <- c(1, 2, 3, 4, 5)
v
typeof(v)
length(v)
v[1]  # Use brackets to access a vector element
v[3]
v <- c(v, 6, 7, 8)  # use c to add items
v <- c(1, 2, 3, "4", "5", "6")
v  # You cannot mix types in a vector
v <- c(x = 1, y = 2, z = 3)  # Named vector
v
v['y']

files <- list.files(path = "data_raw")
files[1]  # Character vector
files[2]
length(files)
```

```
[1] 1 2 3 4 5

[1] "double"

[1] 5

[1] 1

[1] 3

[1] "1" "2" "3" "4" "5" "6"

x y z
1 2 3

y
2

[1] "vocab12.5.csv"

[1] "vocab12.csv"

[1] 24
```

Lists

```
lt <- list(x = 1, y = "3", z = 1:5)
```

Lists

```
lt <- list(x = 1, y = "3", z = 1:5)
lt
```

```
$x
[1] 1

$y
[1] "3"

$z
[1] 1 2 3 4 5
```

Lists

```
lt <- list(x = 1, y = "3", z = 1:5)
lt
lt$z # Access named list elements with $  # Access named list elements wi
```

```
$x
[1] 1

$y
[1] "3"

$z
[1] 1 2 3 4 5

[1] 1 2 3 4 5
```

Lists

```r
lt <- list(x = 1, y = "3", z = 1:5)
lt
lt$z # Access named list elements with $  # Access named list elements wi
lt[[1]]  # Use double brackets to get list elements
```

```
$x
[1] 1

$y
[1] "3"

$z
[1] 1 2 3 4 5

[1] 1 2 3 4 5

[1] 1
```

Lists

```
lt <- list(x = 1, y = "3", z = 1:5)
lt
lt$z # Access named list elements with $  # Access named list elements wi
lt[[1]]  # Use double brackets to get list elements
lt[[3]] <- 5:7
```

```
$x
[1] 1

$y
[1] "3"

$z
[1] 1 2 3 4 5

[1] 1 2 3 4 5

[1] 1
```

Lists

```r
lt <- list(x = 1, y = "3", z = 1:5)
lt
lt$z # Access named list elements with $  # Access named list elements wi
lt[[1]]  # Use double brackets to get list elements
lt[[3]] <- 5:7

# List of lists
lt <- list(lt1 = lt, lt2 = lt, lt3 = lt)
```

```
$x
[1] 1

$y
[1] "3"

$z
[1] 1 2 3 4 5

[1] 1 2 3 4 5

[1] 1
```

Lists

```r
lt <- list(x = 1, y = "3", z = 1:5)
lt
lt$z # Access named list elements with $  # Access named list elements wi
lt[[1]]  # Use double brackets to get list elements
lt[[3]] <- 5:7

# List of lists
lt <- list(lt1 = lt, lt2 = lt, lt3 = lt)
lt$lt1
```

```
$x
[1] 1

$y
[1] "3"

$z
[1] 1 2 3 4 5

[1] 1 2 3 4 5

[1] 1

$x
[1] 1

$y
[1] "3"

$z
[1] 5 6 7
```

Lists

```r
lt <- list(x = 1, y = "3", z = 1:5)
lt
lt$z # Access named list elements with $  # Access named list elements wi
lt[[1]]  # Use double brackets to get list elements
lt[[3]] <- 5:7

# List of lists
lt <- list(lt1 = lt, lt2 = lt, lt3 = lt)
lt$lt1

model <- lm(height ~ mass, data = starwars)
```

```
$x
[1] 1

$y
[1] "3"

$z
[1] 1 2 3 4 5

[1] 1 2 3 4 5

[1] 1

$x
[1] 1

$y
[1] "3"

$z
[1] 5 6 7
```

Lists

```
lt <- list(x = 1, y = "3", z = 1:5)
lt
lt$z # Access named list elements with $  # Access named list elements wi
lt[[1]]  # Use double brackets to get list elements
lt[[3]] <- 5:7

# List of lists
lt <- list(lt1 = lt, lt2 = lt, lt3 = lt)
lt$lt1

model <- lm(height ~ mass, data = starwars)
typeof(model)
```

```
$x
[1] 1

$y
[1] "3"

$z
[1] 1 2 3 4 5

[1] 1 2 3 4 5

[1] 1

$x
[1] 1

$y
[1] "3"

$z
[1] 5 6 7

[1] "list"
```

Lists

```r
lt <- list(x = 1, y = "3", z = 1:5)
lt
lt$z # Access named list elements with $  # Access named list elements wi
lt[[1]]  # Use double brackets to get list elements
lt[[3]] <- 5:7

# List of lists
lt <- list(lt1 = lt, lt2 = lt, lt3 = lt)
lt$lt1

model <- lm(height ~ mass, data = starwars)
typeof(model)
model$coefficients
```

```
$x
[1] 1

$y
[1] "3"

$z
[1] 1 2 3 4 5

[1] 1 2 3 4 5

[1] 1

$x
[1] 1

$y
[1] "3"

$z
[1] 5 6 7

[1] "list"

 (Intercept)         mass
171.68547308   0.02744228
```

Working with Dates and Times

- Correctly parse dates/times from character data
- Calculate differences between dates/times (e.g., ages, task length)
- Determine local time using time zones

Parsing Dates

```
dates
dates %>% mutate(dates1_parsed = ymd(dates1))
```

```
# A tibble: 4 × 3
  dates1     dates2     dates3
  <chr>      <chr>      <chr>
1 2017-09-01 9/01/2018  March 1, 2019
2 2016-09-01 09/01/2017 May 1, 2018
3 2015-09-01 9/1/2016   January 20, 2017
4 2014-09-01 9/1/2015   February 4, 2020

# A tibble: 4 × 4
  dates1     dates2     dates3           dates1_parsed
  <chr>      <chr>      <chr>            <date>
1 2017-09-01 9/01/2018  March 1, 2019    2017-09-01
2 2016-09-01 09/01/2017 May 1, 2018      2016-09-01
3 2015-09-01 9/1/2016   January 20, 2017 2015-09-01
4 2014-09-01 9/1/2015   February 4, 2020 2014-09-01
```

Parsing Dates

```
dates
dates %>% mutate(dates2_parsed = mdy(dates2))
```

```
# A tibble: 4 × 3
  dates1     dates2     dates3
  <chr>      <chr>      <chr>
1 2017-09-01 9/01/2018  March 1, 2019
2 2016-09-01 09/01/2017 May 1, 2018
3 2015-09-01 9/1/2016   January 20, 2017
4 2014-09-01 9/1/2015   February 4, 2020

# A tibble: 4 × 4
  dates1     dates2     dates3           dates2_parsed
  <chr>      <chr>      <chr>            <date>
1 2017-09-01 9/01/2018  March 1, 2019    2018-09-01
2 2016-09-01 09/01/2017 May 1, 2018      2017-09-01
3 2015-09-01 9/1/2016   January 20, 2017 2016-09-01
4 2014-09-01 9/1/2015   February 4, 2020 2015-09-01
```

Parsing Dates

```
dates
dates %>% mutate(dates3_parsed = parse_date(dates3, format = "%B %d, %Y"))
```

```
# A tibble: 4 × 3
  dates1     dates2     dates3
  <chr>      <chr>      <chr>
1 2017-09-01 9/01/2018  March 1, 2019
2 2016-09-01 09/01/2017 May 1, 2018
3 2015-09-01 9/1/2016   January 20, 2017
4 2014-09-01 9/1/2015   February 4, 2020

# A tibble: 4 × 4
  dates1     dates2     dates3           dates3_parsed
  <chr>      <chr>      <chr>            <date>
1 2017-09-01 9/01/2018  March 1, 2019    2019-03-01
2 2016-09-01 09/01/2017 May 1, 2018      2018-05-01
3 2015-09-01 9/1/2016   January 20, 2017 2017-01-20
4 2014-09-01 9/1/2015   February 4, 2020 2020-02-04
```

Time is Weird

- Leap days and leap seconds mean that physical time is not the same as calendar time
  - lubridate `durations` represent physical time
  - lubridate `periods` represent the passage of calendar time

```
time_length(mdy("05-27-2022")-mdy("05-27-1983"), "years")
```

```
[1] 39.00068
```

```
time_length(as.period(interval(mdy("05-27-1983"), mdy("05-27-2022"))), "years")
```

```
[1] 39
```

```
time_length(as.period(interval(mdy("05-27-1983"), mdy("05-28-2022"))), "years")
```

```
[1] 39.00274
```

Age math

```
birth_date <- ymd("2001-09-01", "2000-09-01")
```

Age math

```
birth_date <- ymd("2001-09-01", "2000-09-01")
test_date <- ymd("2020-09-01","2019-09-01")
```

Age math

```
birth_date <- ymd("2001-09-01", "2000-09-01")
test_date <- ymd("2020-09-01","2019-09-01")
(test_date-birth_date)/365.25
```

```
Time differences in days
[1] 19.00068 18.99795
```

Age math

```
birth_date <- ymd("2001-09-01", "2000-09-01")
test_date <- ymd("2020-09-01","2019-09-01")
(test_date-birth_date)/365.25
as.period(interval(start = birth_date, end = test_date), unit = "years")
```

```
Time differences in days
[1] 19.00068 18.99795
```

```
[1] "19y 0m 0d 0H 0M 0S" "19y 0m 0d 0H 0M 0S"
```

Age math

```
birth_date <- ymd("2001-09-01", "2000-09-01")
test_date <- ymd("2020-09-01","2019-09-01")
(test_date-birth_date)/365.25
as.period(interval(start = birth_date, end = test_date), unit = "years")
test_date - years(1) + days(7)
```

```
Time differences in days
[1] 19.00068 18.99795
```

```
[1] "19y 0m 0d 0H 0M 0S" "19y 0m 0d 0H 0M 0S"
```

```
[1] "2019-09-08" "2018-09-08"
```

Time Zones

```
t1 <- as_datetime("2022-01-27 10:39:28", tz = "America/Los_Angeles")
```

Time Zones

```
t1 <- as_datetime("2022-01-27 10:39:28", tz = "America/Los_Angeles")
t1
```

```
[1] "2022-01-27 10:39:28 PST"
```

Time Zones

```
t1 <- as_datetime("2022-01-27 10:39:28", tz = "America/Los_Angeles")
t1
tz(t1)
```

```
[1] "2022-01-27 10:39:28 PST"

[1] "America/Los_Angeles"
```

Time Zones

```
t1 <- as_datetime("2022-01-27 10:39:28", tz = "America/Los_Angeles")
t1
tz(t1)
t2 <- as_datetime("2022-01-27 13:39:28", tz = "America/New_York")
```

```
[1] "2022-01-27 10:39:28 PST"

[1] "America/Los_Angeles"
```

Time Zones

```
t1 <- as_datetime("2022-01-27 10:39:28", tz = "America/Los_Angeles")
t1
tz(t1)
t2 <- as_datetime("2022-01-27 13:39:28", tz = "America/New_York")
t2-t1
```

```
[1] "2022-01-27 10:39:28 PST"

[1] "America/Los_Angeles"

Time difference of 0 secs
```

Time Zones

```
t1 <- as_datetime("2022-01-27 10:39:28", tz = "America/Los_Angeles")
t1
tz(t1)
t2 <- as_datetime("2022-01-27 13:39:28", tz = "America/New_York")
t2-t1
t3 <- as_datetime("2021-10-15 10:39:28", tz = "America/Los_Angeles")
```

```
[1] "2022-01-27 10:39:28 PST"

[1] "America/Los_Angeles"

Time difference of 0 secs
```

Time Zones

```
t1 <- as_datetime("2022-01-27 10:39:28", tz = "America/Los_Angeles")
t1
tz(t1)
t2 <- as_datetime("2022-01-27 13:39:28", tz = "America/New_York")
t2-t1
t3 <- as_datetime("2021-10-15 10:39:28", tz = "America/Los_Angeles")
t3
```

```
[1] "2022-01-27 10:39:28 PST"

[1] "America/Los_Angeles"

Time difference of 0 secs

[1] "2021-10-15 10:39:28 PDT"
```

Time Zones

```
t1 <- as_datetime("2022-01-27 10:39:28", tz = "America/Los_Angeles")
t1
tz(t1)
t2 <- as_datetime("2022-01-27 13:39:28", tz = "America/New_York")
t2-t1
t3 <- as_datetime("2021-10-15 10:39:28", tz = "America/Los_Angeles")
t3
t4 <- as_datetime(now())
```

```
[1] "2022-01-27 10:39:28 PST"

[1] "America/Los_Angeles"

Time difference of 0 secs

[1] "2021-10-15 10:39:28 PDT"
```

Time Zones

```
t1 <- as_datetime("2022-01-27 10:39:28", tz = "America/Los_Angeles")
t1
tz(t1)
t2 <- as_datetime("2022-01-27 13:39:28", tz = "America/New_York")
t2-t1
t3 <- as_datetime("2021-10-15 10:39:28", tz = "America/Los_Angeles")
t3
t4 <- as_datetime(now())
t4
```

```
[1] "2022-01-27 10:39:28 PST"

[1] "America/Los_Angeles"

Time difference of 0 secs

[1] "2021-10-15 10:39:28 PDT"

[1] "2026-10-19 12:27:26 PDT"
```

Time Zones

```
t1 <- as_datetime("2022-01-27 10:39:28", tz = "America/Los_Angeles")
t1
tz(t1)
t2 <- as_datetime("2022-01-27 13:39:28", tz = "America/New_York")
t2-t1
t3 <- as_datetime("2021-10-15 10:39:28", tz = "America/Los_Angeles")
t3
t4 <- as_datetime(now())
t4
as.period(interval(start = t1, end = t4), unit = "days")
```

```
[1] "2022-01-27 10:39:28 PST"

[1] "America/Los_Angeles"

Time difference of 0 secs

[1] "2021-10-15 10:39:28 PDT"

[1] "2026-10-19 12:27:26 PDT"

[1] "1726d 0H 47M 58.8691999912262S"
```

Time Zones

```
t1 <- as_datetime("2022-01-27 10:39:28", tz = "America/Los_Angeles")
t1
tz(t1)
t2 <- as_datetime("2022-01-27 13:39:28", tz = "America/New_York")
t2-t1
t3 <- as_datetime("2021-10-15 10:39:28", tz = "America/Los_Angeles")
t3
t4 <- as_datetime(now())
t4
as.period(interval(start = t1, end = t4), unit = "days")
with_tz(t4, "America/New_York")
```

```
[1] "2022-01-27 10:39:28 PST"

[1] "America/Los_Angeles"

Time difference of 0 secs

[1] "2021-10-15 10:39:28 PDT"

[1] "2026-10-19 12:27:27 PDT"

[1] "1726d 0H 47M 59.0142838954926S"

[1] "2026-10-19 15:27:27 EDT"
```

Time Zones

```
t1 <- as_datetime("2022-01-27 10:39:28", tz = "America/Los_Angeles")
t1
tz(t1)
t2 <- as_datetime("2022-01-27 13:39:28", tz = "America/New_York")
t2-t1
t3 <- as_datetime("2021-10-15 10:39:28", tz = "America/Los_Angeles")
t3
t4 <- as_datetime(now())
t4
as.period(interval(start = t1, end = t4), unit = "days")
with_tz(t4, "America/New_York")
force_tz(t4, "America/New_York")
```

```
[1] "2022-01-27 10:39:28 PST"

[1] "America/Los_Angeles"

Time difference of 0 secs

[1] "2021-10-15 10:39:28 PDT"

[1] "2026-10-19 12:27:27 PDT"

[1] "1726d 0H 47M 59.1997010707855S"

[1] "2026-10-19 15:27:27 EDT"

[1] "2026-10-19 12:27:27 EDT"
```

Factors Represent Categories

```
x <- c(1, 1, 2, 3, 1, 4)
x <- factor(x, levels = c(1,2,3), labels = c("rarely", "neutral", "frequently"))
x
```

```
[1] rarely     rarely     neutral    frequently rarely     <NA>
Levels: rarely neutral frequently
```

- levels restrict the possible set of values
- levels are ordered, which carries forward to output, modeling, graphics, etc.
- factors work as dummy codes; use as.numeric(factor) to treat as a continuous variable in models

Factors

```
ds <- starwars
```

Factors

```
ds <- starwars
unique(ds$eye_color)
```

```
 [1] "blue"           "yellow"       "red"          "brown
 [5] "blue-gray"      "black"        "orange"       "hazel
 [9] "pink"           "unknown"      "red, blue"    "gold"
[13] "green, yellow"  "white"        "dark"
```

Factors

```
ds <- starwars
unique(ds$eye_color)
ds <- ds %>% mutate(eye_f = factor(eye_color))
```

```
 [1] "blue"           "yellow"       "red"          "brown
 [5] "blue-gray"      "black"        "orange"       "hazel
 [9] "pink"           "unknown"      "red, blue"    "gold"
[13] "green, yellow"  "white"        "dark"
```

Factors

```
ds <- starwars
unique(ds$eye_color)
ds <- ds %>% mutate(eye_f = factor(eye_color))
fct_count(ds$eye_f, sort = T) %>% slice_head(n = 5)
```

```
 [1] "blue"           "yellow"       "red"          "brown
 [5] "blue-gray"      "black"        "orange"       "hazel
 [9] "pink"           "unknown"      "red, blue"    "gold"
[13] "green, yellow"  "white"        "dark"
```

```
# A tibble: 5 × 2
  f           n
  <fct>   <int>
1 brown      21
2 blue       19
3 yellow     11
4 black      10
5 orange      8
```

Factors

```
ds <- starwars
unique(ds$eye_color)
ds <- ds %>% mutate(eye_f = factor(eye_color))
fct_count(ds$eye_f, sort = T) %>% slice_head(n = 5)
fct_count(ds$eye_f, prop = T) %>% slice_head(n = 5)
```

```
 [1] "blue"          "yellow"        "red"           "brown
 [5] "blue-gray"     "black"         "orange"        "hazel
 [9] "pink"          "unknown"       "red, blue"     "gold"
[13] "green, yellow" "white"         "dark"

# A tibble: 5 × 2
  f           n
  <fct>   <int>
1 brown      21
2 blue       19
3 yellow     11
4 black      10
5 orange      8

# A tibble: 5 × 3
  f              n       p
  <fct>      <int>   <dbl>
1 black         10 0.115
2 blue          19 0.218
3 blue-gray      1 0.0115
4 brown         21 0.241
5 dark           1 0.0115
```

Setting factor levels

```
ds <- ds %>% select(name, eye_f)
ds
```

```
# A tibble: 87 × 2
   name               eye_f
   <chr>              <fct>
 1 Luke Skywalker     blue
 2 C-3PO              yellow
 3 R2-D2              red
 4 Darth Vader        yellow
 5 Leia Organa        brown
 6 Owen Lars          blue
 7 Beru Whitesun Lars blue
 8 R5-D4              red
 9 Biggs Darklighter  brown
10 Obi-Wan Kenobi     blue-gray
# ℹ 77 more rows
```

Setting factor levels

```
ds <- ds %>% select(name, eye_f)
ds %>% mutate(eye_f2 = factor(eye_f, levels = c("blue", "brown")))
```

```
# A tibble: 87 × 3
   name               eye_f     eye_f2
   <chr>              <fct>     <fct>
 1 Luke Skywalker     blue      blue
 2 C-3PO              yellow    <NA>
 3 R2-D2              red       <NA>
 4 Darth Vader        yellow    <NA>
 5 Leia Organa        brown     brown
 6 Owen Lars          blue      blue
 7 Beru Whitesun Lars blue      blue
 8 R5-D4              red       <NA>
 9 Biggs Darklighter  brown     brown
10 Obi-Wan Kenobi     blue-gray <NA>
# ℹ 77 more rows
```

Setting factor levels

```
ds <- ds %>% select(name, eye_f)
ds %>% mutate(eye_f2 = factor(eye_f, levels = c("blue", "brown"), labels = c("bl","br")))
```

```
# A tibble: 87 × 3
   name               eye_f     eye_f2
   <chr>              <fct>     <fct>
 1 Luke Skywalker     blue      bl
 2 C-3PO              yellow    <NA>
 3 R2-D2              red       <NA>
 4 Darth Vader        yellow    <NA>
 5 Leia Organa        brown     br
 6 Owen Lars          blue      bl
 7 Beru Whitesun Lars blue      bl
 8 R5-D4              red       <NA>
 9 Biggs Darklighter  brown     br
10 Obi-Wan Kenobi     blue-gray <NA>
# ℹ 77 more rows
```

Setting factor levels

```
ds <- ds %>% select(name, eye_f)
ds %>% mutate(eye_f2 = fct_lump(eye_f, n = 3))
```

```
# A tibble: 87 × 3
   name               eye_f     eye_f2
   <chr>              <fct>     <fct>
 1 Luke Skywalker     blue      blue
 2 C-3PO              yellow    yellow
 3 R2-D2              red       Other
 4 Darth Vader        yellow    yellow
 5 Leia Organa        brown     brown
 6 Owen Lars          blue      blue
 7 Beru Whitesun Lars blue      blue
 8 R5-D4              red       Other
 9 Biggs Darklighter  brown     brown
10 Obi-Wan Kenobi     blue-gray Other
# ℹ 77 more rows
```

Setting factor levels

```
ds <- ds %>% select(name, eye_f)
ds %>% mutate(eye_type = fct_collapse(eye_f, human = c("blue", "brown"), nonhuman = c("yellow", "r
```

```
# A tibble: 87 × 3
   name               eye_f     eye_type
   <chr>              <fct>     <fct>
 1 Luke Skywalker     blue      human
 2 C-3PO              yellow    nonhuman
 3 R2-D2              red       nonhuman
 4 Darth Vader        yellow    nonhuman
 5 Leia Organa        brown     human
 6 Owen Lars          blue      human
 7 Beru Whitesun Lars blue      human
 8 R5-D4              red       nonhuman
 9 Biggs Darklighter  brown     human
10 Obi-Wan Kenobi     blue-gray blue-gray
# ℹ 77 more rows
```

Reordering factor levels

```
fct_count(ds$eye_f)
```

```
# A tibble: 15 × 2
   f                 n
   <fct>         <int>
 1 black            10
 2 blue             19
 3 blue-gray         1
 4 brown            21
 5 dark              1
 6 gold              1
 7 green, yellow     1
 8 hazel             3
 9 orange            8
10 pink              1
11 red               5
12 red, blue         1
13 unknown           3
14 white             1
15 yellow           11
```

Reordering factor levels

```
fct_count(fct_rev(ds$eye_f))
```

```
# A tibble: 15 × 2
   f                 n
   <fct>         <int>
 1 yellow           11
 2 white             1
 3 unknown           3
 4 red, blue         1
 5 red               5
 6 pink              1
 7 orange            8
 8 hazel             3
 9 green, yellow     1
10 gold              1
11 dark              1
12 brown            21
13 blue-gray         1
14 blue             19
15 black            10
```

Reordering factor levels

```
fct_count(fct_infreq(ds$eye_f))
```

```
# A tibble: 15 × 2
   f                 n
   <fct>         <int>
 1 brown            21
 2 blue             19
 3 yellow           11
 4 black            10
 5 orange            8
 6 red               5
 7 hazel             3
 8 unknown           3
 9 blue-gray         1
10 dark              1
11 gold              1
12 green, yellow     1
13 pink              1
14 red, blue         1
15 white             1
```

Reordering factor levels

```
fct_count(factor(ds$eye_f, levels = c("red", "brown", "blue")))
```

```
# A tibble: 4 × 2
  f         n
  <fct> <int>
1 red       5
2 brown    21
3 blue     19
4 <NA>     42
```

Part 2: Tidying Data

Why do data need to be wrangled and tidied?

- Data files are often created without the foresight of how they might be used by computers
    - Instead, they are made to look nice for humans
- We need to tidy data to make it convenient for programming
    - Format is dictated by analyses
    - You may need multiple formats

Tidy Data (Wickham, 2014)

- Each variable is a column
  - Variable: “All values that measure the same underlying attribute”
- Each observation is a row
  - Observation: Unit for which all variables were measured (person, session, trial, etc.)
- Each type of observational unit is a table
  - e.g., participant questionnaire vs trial-by-trial data

Dataset Preparations

```
print(relig_income, n = 5)
relig_income %>% select(religion, contains("$"))
```

```
# A tibble: 18 × 11
  religion  `<$10k` `$10-20k` `$20-30k` `$30-40k` `$40-50k` `$50-75k` `$75-100k`
  <chr>       <dbl>     <dbl>     <dbl>     <dbl>     <dbl>     <dbl>      <dbl>
1 Agnostic       27        34        60        81        76       137        122
2 Atheist        12        27        37        52        35        70         73
3 Buddhist       27        21        30        34        33        58         62
4 Catholic      418       617       732       670       638      1116        949
5 Don't kn…      15        14        15        11        10        35         21
# ℹ 13 more rows
# ℹ 3 more variables: `$100-150k` <dbl>, `>150k` <dbl>,
#   `Don't know/refused` <dbl>
```

```
# A tibble: 18 × 9
   religion `<$10k` `$10-20k` `$20-30k` `$30-40k` `$40-50k` `$50-75k` `$75-100k`
   <chr>      <dbl>     <dbl>     <dbl>     <dbl>     <dbl>     <dbl>      <dbl>
 1 Agnostic      27        34        60        81        76       137        122
 2 Atheist       12        27        37        52        35        70         73
 3 Buddhist      27        21        30        34        33        58         62
 4 Catholic     418       617       732       670       638      1116        949
 5 Don't k…      15        14        15        11        10        35         21
 6 Evangel…     575       869      1064       982       881      1486        949
 7 Hindu          1         9         7         9        11        34         47
 8 Histori…     228       244       236       238       197       223        131
 9 Jehovah…      20        27        24        24        21        30         15
10 Jewish        19        19        25        25        30        95         69
11 Mainlin…     289       495       619       655       651      1107        939
12 Mormon        29        40        48        51        56       112         85
13 Muslim         6         7         9        10         9        23         16
```

Dataset Preparations

```
print(relig_income, n = 5)
relig_income %>% select(religion, starts_with("$"))
```

```
# A tibble: 18 × 11
  religion  `<$10k` `$10-20k` `$20-30k` `$30-40k` `$40-50k` `$50-75k` `$75-100k`
  <chr>       <dbl>     <dbl>     <dbl>     <dbl>     <dbl>     <dbl>      <dbl>
1 Agnostic       27        34        60        81        76       137        122
2 Atheist        12        27        37        52        35        70         73
3 Buddhist       27        21        30        34        33        58         62
4 Catholic      418       617       732       670       638      1116        949
5 Don't kn…      15        14        15        11        10        35         21
# ℹ 13 more rows
# ℹ 3 more variables: `$100-150k` <dbl>, `>150k` <dbl>,
#   `Don't know/refused` <dbl>

# A tibble: 18 × 8
   religion         `$10-20k` `$20-30k` `$30-40k` `$40-50k` `$50-75k` `$75-100k`
   <chr>                <dbl>     <dbl>     <dbl>     <dbl>     <dbl>      <dbl>
 1 Agnostic                34        60        81        76       137        122
 2 Atheist                 27        37        52        35        70         73
 3 Buddhist                21        30        34        33        58         62
 4 Catholic               617       732       670       638      1116        949
 5 Don't know/refu…        14        15        11        10        35         21
 6 Evangelical Prot       869      1064       982       881      1486        949
 7 Hindu                    9         7         9        11        34         47
 8 Historically Bl…       244       236       238       197       223        131
 9 Jehovah's Witne…        27        24        24        21        30         15
10 Jewish                  19        25        25        30        95         69
11 Mainline Prot          495       619       655       651      1107        939
12 Mormon                  40        48        51        56       112         85
13 Muslim                   7         9        10         9        23         16
```

Dataset Preparations

```
print(relig_income, n = 5)
relig_income %>% select(religion, contains("k"))
```

```
# A tibble: 18 × 11
  religion  `<$10k` `$10-20k` `$20-30k` `$30-40k` `$40-50k` `$50-75k` `$75-100k`
  <chr>       <dbl>     <dbl>     <dbl>     <dbl>     <dbl>     <dbl>      <dbl>
1 Agnostic       27        34        60        81        76       137        122
2 Atheist        12        27        37        52        35        70         73
3 Buddhist       27        21        30        34        33        58         62
4 Catholic      418       617       732       670       638      1116        949
5 Don't kn…      15        14        15        11        10        35         21
# ℹ 13 more rows
# ℹ 3 more variables: `$100-150k` <dbl>, `>150k` <dbl>,
#   `Don't know/refused` <dbl>
```

```
# A tibble: 18 × 11
   religion `<$10k` `$10-20k` `$20-30k` `$30-40k` `$40-50k` `$50-75k` `$75-100k`
   <chr>      <dbl>     <dbl>     <dbl>     <dbl>     <dbl>     <dbl>      <dbl>
 1 Agnostic      27        34        60        81        76       137        122
 2 Atheist       12        27        37        52        35        70         73
 3 Buddhist      27        21        30        34        33        58         62
 4 Catholic     418       617       732       670       638      1116        949
 5 Don't k…      15        14        15        11        10        35         21
 6 Evangel…     575       869      1064       982       881      1486        949
 7 Hindu          1         9         7         9        11        34         47
 8 Histori…     228       244       236       238       197       223        131
 9 Jehovah…      20        27        24        24        21        30         15
10 Jewish        19        19        25        25        30        95         69
11 Mainlin…     289       495       619       655       651      1107        939
12 Mormon        29        40        48        51        56       112         85
13 Muslim         6         7         9        10         9        23         16
```

Dataset Preparations

```
print(relig_income, n = 5)
relig_income %>% select(religion, starts_with("$"), ends_with("k"))
```

```
# A tibble: 18 × 11
  religion  `<$10k` `$10-20k` `$20-30k` `$30-40k` `$40-50k` `$50-75k` `$75-100k`
  <chr>       <dbl>     <dbl>     <dbl>     <dbl>     <dbl>     <dbl>      <dbl>
1 Agnostic       27        34        60        81        76       137        122
2 Atheist        12        27        37        52        35        70         73
3 Buddhist       27        21        30        34        33        58         62
4 Catholic      418       617       732       670       638      1116        949
5 Don't kn…      15        14        15        11        10        35         21
# ℹ 13 more rows
# ℹ 3 more variables: `$100-150k` <dbl>, `>150k` <dbl>,
#   `Don't know/refused` <dbl>

# A tibble: 18 × 10
   religion          `$10-20k` `$20-30k` `$30-40k` `$40-50k` `$50-75k` `$75-100k`
   <chr>                 <dbl>     <dbl>     <dbl>     <dbl>     <dbl>      <dbl>
 1 Agnostic                 34        60        81        76       137        122
 2 Atheist                  27        37        52        35        70         73
 3 Buddhist                 21        30        34        33        58         62
 4 Catholic                617       732       670       638      1116        949
 5 Don't know/refu…         14        15        11        10        35         21
 6 Evangelical Prot        869      1064       982       881      1486        949
 7 Hindu                     9         7         9        11        34         47
 8 Historically Bl…        244       236       238       197       223        131
 9 Jehovah's Witne…         27        24        24        21        30         15
10 Jewish                   19        25        25        30        95         69
11 Mainline Prot           495       619       655       651      1107        939
12 Mormon                   40        48        51        56       112         85
13 Muslim                    7         9        10         9        23         16
```

Dataset Transformations

Multiple Columns -> One Variable

```
ds <- read_csv('data_cleaned/vocab_annoying_headers.csv')
```

Multiple Columns -> One Variable

```
ds <- read_csv('data_cleaned/vocab_annoying_headers.csv') %>%
  mutate(id = "Jonah", dob = as.Date("2017-08-30"), .before = "age_12")
```

Multiple Columns -> One Variable

```
ds <- read_csv('data_cleaned/vocab_annoying_headers.csv') %>%
  mutate(id = "Jonah", dob = as.Date("2017-08-30"), .before = "age_12")
ds %>% print(n = 25)
```

```
# A tibble: 51 × 27
   id    dob        age_12  age_12.5 age_
   <chr> <date>     <chr>   <chr>    <lgl
 1 Jonah 2017-08-30 book    dad      NA
 2 Jonah 2017-08-30 ball    <NA>     NA
 3 Jonah 2017-08-30 bye bye <NA>     NA
 4 Jonah 2017-08-30 <NA>    <NA>     NA
 5 Jonah 2017-08-30 <NA>    <NA>     NA
 6 Jonah 2017-08-30 <NA>    <NA>     NA
 7 Jonah 2017-08-30 <NA>    <NA>     NA
 8 Jonah 2017-08-30 <NA>    <NA>     NA
 9 Jonah 2017-08-30 <NA>    <NA>     NA
10 Jonah 2017-08-30 <NA>    <NA>     NA
11 Jonah 2017-08-30 <NA>    <NA>     NA
12 Jonah 2017-08-30 <NA>    <NA>     NA
13 Jonah 2017-08-30 <NA>    <NA>     NA
14 Jonah 2017-08-30 <NA>    <NA>     NA
15 Jonah 2017-08-30 <NA>    <NA>     NA
16 Jonah 2017-08-30 <NA>    <NA>     NA
17 Jonah 2017-08-30 <NA>    <NA>     NA
18 Jonah 2017-08-30 <NA>    <NA>     NA
19 Jonah 2017-08-30 <NA>    <NA>     NA
20 Jonah 2017-08-30 <NA>    <NA>     NA
21 Jonah 2017-08-30 <NA>    <NA>     NA
22 Jonah 2017-08-30 <NA>    <NA>     NA
23 Jonah 2017-08-30 <NA>    <NA>     NA
24 Jonah 2017-08-30 <NA>    <NA>     NA
25 Jonah 2017-08-30 <NA>    <NA>     NA
# ℹ 26 more rows
# ℹ 18 more variables: age_15.5 <chr>, ag
#   age_17 <chr>, age_17.5 <chr>, age_18
#   age_19.5 <chr>, age_20 <chr>, age_20.
```

Multiple Columns -> One Variable

```
ds <- read_csv('data_cleaned/vocab_annoying_headers.csv') %>%
  mutate(id = "Jonah", dob = as.Date("2017-08-30"), .before = "age_12")
ds %>% print(n = 25)
ds %>% pivot_longer(-(id:dob), names_to = "age", values_to = "word")
```

```
# A tibble: 51 × 27
   id    dob        age_12  age_12.5 age_
   <chr> <date>     <chr>   <chr>    <lgl
 1 Jonah 2017-08-30 book    dad      NA
 2 Jonah 2017-08-30 ball    <NA>     NA
 3 Jonah 2017-08-30 bye bye <NA>     NA
 4 Jonah 2017-08-30 <NA>    <NA>     NA
 5 Jonah 2017-08-30 <NA>    <NA>     NA
 6 Jonah 2017-08-30 <NA>    <NA>     NA
 7 Jonah 2017-08-30 <NA>    <NA>     NA
 8 Jonah 2017-08-30 <NA>    <NA>     NA
 9 Jonah 2017-08-30 <NA>    <NA>     NA
10 Jonah 2017-08-30 <NA>    <NA>     NA
11 Jonah 2017-08-30 <NA>    <NA>     NA
12 Jonah 2017-08-30 <NA>    <NA>     NA
13 Jonah 2017-08-30 <NA>    <NA>     NA
14 Jonah 2017-08-30 <NA>    <NA>     NA
15 Jonah 2017-08-30 <NA>    <NA>     NA
16 Jonah 2017-08-30 <NA>    <NA>     NA
17 Jonah 2017-08-30 <NA>    <NA>     NA
18 Jonah 2017-08-30 <NA>    <NA>     NA
19 Jonah 2017-08-30 <NA>    <NA>     NA
20 Jonah 2017-08-30 <NA>    <NA>     NA
21 Jonah 2017-08-30 <NA>    <NA>     NA
22 Jonah 2017-08-30 <NA>    <NA>     NA
23 Jonah 2017-08-30 <NA>    <NA>     NA
24 Jonah 2017-08-30 <NA>    <NA>     NA
25 Jonah 2017-08-30 <NA>    <NA>     NA
# ℹ 26 more rows
# ℹ 18 more variables: age_15.5 <chr>, ag
#   age_17 <chr>, age_17.5 <chr>, age_18
#   age_19.5 <chr>, age_20 <chr>, age_20.
```

Multiple Columns -> One Variable

```
ds <- read_csv('data_cleaned/vocab_annoying_headers.csv') %>%
  mutate(id = "Jonah", dob = as.Date("2017-08-30"), .before = "age_12")
ds %>% print(n = 25)
ds %>% pivot_longer(-(id:dob), names_to = "age", values_to = "word")
ds %>% pivot_longer(-(id:dob), names_to = "age",
                    names_prefix = "age_", values_to = "word")
```

```
# A tibble: 51 × 27
   id    dob        age_12  age_12.5 age_
   <chr> <date>     <chr>   <chr>    <lgl
 1 Jonah 2017-08-30 book    dad      NA
 2 Jonah 2017-08-30 ball    <NA>     NA
 3 Jonah 2017-08-30 bye bye <NA>     NA
 4 Jonah 2017-08-30 <NA>    <NA>     NA
 5 Jonah 2017-08-30 <NA>    <NA>     NA
 6 Jonah 2017-08-30 <NA>    <NA>     NA
 7 Jonah 2017-08-30 <NA>    <NA>     NA
 8 Jonah 2017-08-30 <NA>    <NA>     NA
 9 Jonah 2017-08-30 <NA>    <NA>     NA
10 Jonah 2017-08-30 <NA>    <NA>     NA
11 Jonah 2017-08-30 <NA>    <NA>     NA
12 Jonah 2017-08-30 <NA>    <NA>     NA
13 Jonah 2017-08-30 <NA>    <NA>     NA
14 Jonah 2017-08-30 <NA>    <NA>     NA
15 Jonah 2017-08-30 <NA>    <NA>     NA
16 Jonah 2017-08-30 <NA>    <NA>     NA
17 Jonah 2017-08-30 <NA>    <NA>     NA
18 Jonah 2017-08-30 <NA>    <NA>     NA
19 Jonah 2017-08-30 <NA>    <NA>     NA
20 Jonah 2017-08-30 <NA>    <NA>     NA
21 Jonah 2017-08-30 <NA>    <NA>     NA
22 Jonah 2017-08-30 <NA>    <NA>     NA
23 Jonah 2017-08-30 <NA>    <NA>     NA
24 Jonah 2017-08-30 <NA>    <NA>     NA
25 Jonah 2017-08-30 <NA>    <NA>     NA
# ℹ 26 more rows
# ℹ 18 more variables: age_15.5 <chr>, ag
#   age_17 <chr>, age_17.5 <chr>, age_18
#   age_19.5 <chr>, age_20 <chr>, age_20.
```

Multiple Columns -> One Variable

```
ds <- read_csv('data_cleaned/vocab_annoying_headers.csv') %>%
  mutate(id = "Jonah", dob = as.Date("2017-08-30"), .before = "age_12")
ds %>% print(n = 25)
ds %>% pivot_longer(-(id:dob), names_to = "age", values_to = "word")
ds %>% pivot_longer(-(id:dob), names_to = "age",
                    names_prefix = "age_", values_to = "word")
ds %>% pivot_longer(-(id:dob), names_to = "age",
                    values_to = "word", values_drop_na = T, names_prefix
```

```
# A tibble: 51 × 27
   id    dob        age_12  age_12.5 age_
   <chr> <date>     <chr>   <chr>    <lgl
 1 Jonah 2017-08-30 book    dad      NA
 2 Jonah 2017-08-30 ball    <NA>     NA
 3 Jonah 2017-08-30 bye bye <NA>     NA
 4 Jonah 2017-08-30 <NA>    <NA>     NA
 5 Jonah 2017-08-30 <NA>    <NA>     NA
 6 Jonah 2017-08-30 <NA>    <NA>     NA
 7 Jonah 2017-08-30 <NA>    <NA>     NA
 8 Jonah 2017-08-30 <NA>    <NA>     NA
 9 Jonah 2017-08-30 <NA>    <NA>     NA
10 Jonah 2017-08-30 <NA>    <NA>     NA
11 Jonah 2017-08-30 <NA>    <NA>     NA
12 Jonah 2017-08-30 <NA>    <NA>     NA
13 Jonah 2017-08-30 <NA>    <NA>     NA
14 Jonah 2017-08-30 <NA>    <NA>     NA
15 Jonah 2017-08-30 <NA>    <NA>     NA
16 Jonah 2017-08-30 <NA>    <NA>     NA
17 Jonah 2017-08-30 <NA>    <NA>     NA
18 Jonah 2017-08-30 <NA>    <NA>     NA
19 Jonah 2017-08-30 <NA>    <NA>     NA
20 Jonah 2017-08-30 <NA>    <NA>     NA
21 Jonah 2017-08-30 <NA>    <NA>     NA
22 Jonah 2017-08-30 <NA>    <NA>     NA
23 Jonah 2017-08-30 <NA>    <NA>     NA
24 Jonah 2017-08-30 <NA>    <NA>     NA
25 Jonah 2017-08-30 <NA>    <NA>     NA
# ℹ 26 more rows
# ℹ 18 more variables: age_15.5 <chr>, ag
#   age_17 <chr>, age_17.5 <chr>, age_18
#   age_19.5 <chr>, age_20 <chr>, age_20.
```

One Column -> Multiple variables

```
ds <- read_delim(list.files(path = "data_headers", full
                 col_names = c("field", "value"), col_t
                 delim = " ", n_max = 8, id = "filename
```

One Column -> Multiple variables

```
ds <- read_delim(list.files(path = "data_headers", full
                 col_names = c("field", "value"), col_t
                 delim = " ", n_max = 8, id = "filename
print(ds)
```

```
# A tibble: 96 × 3
   filename                 field       value
   <chr>                    <chr>       <chr>
 1 data_headers/6191_1.txt  Participant 6191
 2 data_headers/6191_1.txt  Age         25
 3 data_headers/6191_1.txt  Sex         male
 4 data_headers/6191_1.txt  Order       1
 5 data_headers/6191_1.txt  FirstSpeed  4
 6 data_headers/6191_1.txt  Block       1
 7 data_headers/6191_1.txt  speed       4
 8 data_headers/6191_1.txt  condition   near
 9 data_headers/6191_2.txt  Participant 6191
10 data_headers/6191_2.txt  Age         25
# ℹ 86 more rows
```

One Column -> Multiple variables

```
ds <- read_delim(list.files(path = "data_headers", full
                 col_names = c("field", "value"), col_t
                 delim = " ", n_max = 8, id = "filename
print(ds)

ds <- ds %>% pivot_wider(id_cols = "filename",
                         names_from = "field",
                         values_from = "value")
```

```
# A tibble: 96 × 3
   filename                 field       value
   <chr>                    <chr>       <chr>
 1 data_headers/6191_1.txt  Participant 6191
 2 data_headers/6191_1.txt  Age         25
 3 data_headers/6191_1.txt  Sex         male
 4 data_headers/6191_1.txt  Order       1
 5 data_headers/6191_1.txt  FirstSpeed  4
 6 data_headers/6191_1.txt  Block       1
 7 data_headers/6191_1.txt  speed       4
 8 data_headers/6191_1.txt  condition   near
 9 data_headers/6191_2.txt  Participant 6191
10 data_headers/6191_2.txt  Age         25
# ℹ 86 more rows
```

One Column -> Multiple variables

```
ds <- read_delim(list.files(path = "data_headers", full
                 col_names = c("field", "value"), col_t
                 delim = " ", n_max = 8, id = "filename
print(ds)

ds <- ds %>% pivot_wider(id_cols = "filename",
                         names_from = "field",
                         values_from = "value")
ds
```

```
# A tibble: 96 × 3
   filename                 field       value
   <chr>                    <chr>       <chr>
 1 data_headers/6191_1.txt  Participant 6191
 2 data_headers/6191_1.txt  Age         25
 3 data_headers/6191_1.txt  Sex         male
 4 data_headers/6191_1.txt  Order       1
 5 data_headers/6191_1.txt  FirstSpeed  4
 6 data_headers/6191_1.txt  Block       1
 7 data_headers/6191_1.txt  speed       4
 8 data_headers/6191_1.txt  condition   near
 9 data_headers/6191_2.txt  Participant 6191
10 data_headers/6191_2.txt  Age         25
# ℹ 86 more rows

# A tibble: 12 × 9
   filename       Participant Age   Sex   Order FirstSpeed
   <chr>          <chr>       <chr> <chr> <chr> <chr>
 1 data_headers/… 6191        25    male  1     4
 2 data_headers/… 6191        25    male  1     4
 3 data_headers/… 6191        25    male  1     4
 4 data_headers/… 6191        25    male  1     4
 5 data_headers/… 6191        25    male  1     4
 6 data_headers/… 6191        25    male  1     4
 7 data_headers/… 6192        21    male  2     4
 8 data_headers/… 6192        21    male  2     4
 9 data_headers/… 6192        21    male  2     4
10 data_headers/… 6192        21    male  2     4
11 data_headers/… 6192        21    male  2     4
12 data_headers/… 6192        21    male  2     4
```

One Variable -> Multiple Variables

```
ds
```

```
# A tibble: 12 × 9
   filename       Participant Age   Sex   Order FirstSpeed
   <chr>          <chr>       <chr> <chr> <chr> <chr>
 1 data_headers/… 6191        25    male  1     4
 2 data_headers/… 6191        25    male  1     4
 3 data_headers/… 6191        25    male  1     4
 4 data_headers/… 6191        25    male  1     4
 5 data_headers/… 6191        25    male  1     4
 6 data_headers/… 6191        25    male  1     4
 7 data_headers/… 6192        21    male  2     4
 8 data_headers/… 6192        21    male  2     4
 9 data_headers/… 6192        21    male  2     4
10 data_headers/… 6192        21    male  2     4
11 data_headers/… 6192        21    male  2     4
12 data_headers/… 6192        21    male  2     4
```

One Variable -> Multiple Variables

```
ds
ds %>% separate_wider_delim(filename, delim = "/", name
```

```
# A tibble: 12 × 9
   filename        Participant Age   Sex   Order FirstSpeed
   <chr>           <chr>       <chr> <chr> <chr> <chr>
 1 data_headers/… 6191        25    male  1     4
 2 data_headers/… 6191        25    male  1     4
 3 data_headers/… 6191        25    male  1     4
 4 data_headers/… 6191        25    male  1     4
 5 data_headers/… 6191        25    male  1     4
 6 data_headers/… 6191        25    male  1     4
 7 data_headers/… 6192        21    male  2     4
 8 data_headers/… 6192        21    male  2     4
 9 data_headers/… 6192        21    male  2     4
10 data_headers/… 6192        21    male  2     4
11 data_headers/… 6192        21    male  2     4
12 data_headers/… 6192        21    male  2     4
```

```
# A tibble: 12 × 10
   dir      file  Participant Age   Sex   Order FirstSpeed
   <chr>    <chr> <chr>       <chr> <chr> <chr> <chr>
 1 data_he… 6191… 6191        25    male  1     4
 2 data_he… 6191… 6191        25    male  1     4
 3 data_he… 6191… 6191        25    male  1     4
 4 data_he… 6191… 6191        25    male  1     4
 5 data_he… 6191… 6191        25    male  1     4
 6 data_he… 6191… 6191        25    male  1     4
 7 data_he… 6192… 6192        21    male  2     4
 8 data_he… 6192… 6192        21    male  2     4
 9 data_he… 6192… 6192        21    male  2     4
10 data_he… 6192… 6192        21    male  2     4
11 data_he… 6192… 6192        21    male  2     4
12 data_he… 6192… 6192        21    male  2     4
```

Multiple Variables -> One Variable

-unite()

Binding Columns

```
star_wars_numeric <- starwars %>% select(where(is.numer
```

Binding Columns

```
star_wars_numeric <- starwars %>% select(where(is.numer
star_wars_numeric
```

```
# A tibble: 87 × 3
   height  mass birth_year
    <int> <dbl>      <dbl>
 1    172    77       19
 2    167    75      112
 3     96    32       33
 4    202   136       41.9
 5    150    49       19
 6    178   120       52
 7    165    75       47
 8     97    32       NA
 9    183    84       24
10    182    77       57
# ℹ 77 more rows
```

Binding Columns

```
star_wars_numeric <- starwars %>% select(where(is.numer
star_wars_numeric
star_wars_char <- starwars %>% select(where(is.characte
```

```
# A tibble: 87 × 3
   height  mass birth_year
    <int> <dbl>      <dbl>
 1    172    77       19
 2    167    75      112
 3     96    32       33
 4    202   136       41.9
 5    150    49       19
 6    178   120       52
 7    165    75       47
 8     97    32       NA
 9    183    84       24
10    182    77       57
# ℹ 77 more rows
```

Binding Columns

```
star_wars_numeric <- starwars %>% select(where(is.numer
star_wars_numeric
star_wars_char <- starwars %>% select(where(is.characte
star_wars_char
```

```
# A tibble: 87 × 3
   height  mass birth_year
    <int> <dbl>      <dbl>
 1    172    77       19
 2    167    75      112
 3     96    32       33
 4    202   136       41.9
 5    150    49       19
 6    178   120       52
 7    165    75       47
 8     97    32       NA
 9    183    84       24
10    182    77       57
# ℹ 77 more rows
```

```
# A tibble: 87 × 8
   name            hair_color skin_color eye_color sex    gen
   <chr>           <chr>      <chr>      <chr>     <chr>  <ch
 1 Luke Skywalker  blond      fair       blue      male   mas
 2 C-3PO           <NA>       gold       yellow    none   mas
 3 R2-D2           <NA>       white, bl… red       none   mas
 4 Darth Vader     none       white      yellow    male   mas
 5 Leia Organa     brown      light      brown     fema…  fem
 6 Owen Lars       brown, gr… light      blue      male   mas
 7 Beru Whitesun…  brown      light      blue      fema…  fem
 8 R5-D4           <NA>       white, red red       none   mas
 9 Biggs Darklig…  black      light      brown     male   mas
10 Obi-Wan Kenobi  auburn, w… fair       blue-gray male   mas
# ℹ 77 more rows
```

Binding Columns

```
star_wars_numeric <- starwars %>% select(where(is.numer
star_wars_numeric
star_wars_char <- starwars %>% select(where(is.characte
star_wars_char
star_wars_combined <- bind_cols(star_wars_numeric, star
```

```
# A tibble: 87 × 3
   height  mass birth_year
    <int> <dbl>      <dbl>
 1    172    77       19
 2    167    75      112
 3     96    32       33
 4    202   136       41.9
 5    150    49       19
 6    178   120       52
 7    165    75       47
 8     97    32       NA
 9    183    84       24
10    182    77       57
# ℹ 77 more rows
```

```
# A tibble: 87 × 8
   name            hair_color skin_color eye_color sex   gen
   <chr>           <chr>      <chr>      <chr>     <chr> <ch
 1 Luke Skywalker  blond      fair       blue      male  mas
 2 C-3PO           <NA>       gold       yellow    none  mas
 3 R2-D2           <NA>       white, bl… red       none  mas
 4 Darth Vader     none       white      yellow    male  mas
 5 Leia Organa     brown      light      brown     fema… fem
 6 Owen Lars       brown, gr… light      blue      male  mas
 7 Beru Whitesun… brown       light      blue      fema… fem
 8 R5-D4           <NA>       white, red red       none  mas
 9 Biggs Darklig… black       light      brown     male  mas
10 Obi-Wan Kenobi auburn, w… fair       blue-gray male  mas
# ℹ 77 more rows
```

Binding Columns

```
star_wars_numeric <- starwars %>% select(where(is.numer
star_wars_numeric
star_wars_char <- starwars %>% select(where(is.characte
star_wars_char
star_wars_combined <- bind_cols(star_wars_numeric, star
star_wars_combined
```

```
# A tibble: 87 × 3
   height  mass birth_year
    <int> <dbl>      <dbl>
 1    172    77       19
 2    167    75      112
 3     96    32       33
 4    202   136       41.9
 5    150    49       19
 6    178   120       52
 7    165    75       47
 8     97    32       NA
 9    183    84       24
10    182    77       57
# ℹ 77 more rows
```

```
# A tibble: 87 × 8
   name           hair_color skin_color eye_color sex   gen
   <chr>          <chr>      <chr>      <chr>     <chr> <ch
 1 Luke Skywalker blond      fair       blue      male  mas
 2 C-3PO          <NA>       gold       yellow    none  mas
 3 R2-D2          <NA>       white, bl… red       none  mas
 4 Darth Vader    none       white      yellow    male  mas
 5 Leia Organa    brown      light      brown     fema… fem
 6 Owen Lars      brown, gr… light      blue      male  mas
 7 Beru Whitesun… brown      light      blue      fema… fem
 8 R5-D4          <NA>       white, red red       none  mas
 9 Biggs Darklig… black      light      brown     male  mas
10 Obi-Wan Kenobi auburn, w… fair       blue-gray male  mas
# ℹ 77 more rows
```

```
# A tibble: 87 × 11
```

Binding Rows

```
s1 <- starwars %>% slice_sample(n = 2)   #2x14
```

Binding Rows

```
s1 <- starwars %>% slice_sample(n = 2)  #2x14
s1
```

```
# A tibble: 2 × 14
  name      height  mass hair_color skin_color eye_color bi
  <chr>      <int> <dbl> <chr>      <chr>      <chr>
1 Obi-Wan …    182    77 auburn, w… fair       blue-gray
2 Ben Quad…    163    65 none       grey, gre… orange
# ℹ 5 more variables: homeworld <chr>, species <chr>, films
#   vehicles <list>, starships <list>
```

Binding Rows

```
s1 <- starwars %>% slice_sample(n = 2)  #2x14
s1
s2 <- starwars %>% slice_sample(n = 2)  #2x14
```

```
# A tibble: 2 × 14
  name      height  mass hair_color skin_color eye_color bi
  <chr>      <int> <dbl> <chr>      <chr>      <chr>
1 Taun We      213    NA none       grey       black
2 Yarael P…    264    NA none       white      yellow
# ℹ 5 more variables: homeworld <chr>, species <chr>, films
#   vehicles <list>, starships <list>
```

Binding Rows

```
s1 <- starwars %>% slice_sample(n = 2)  #2x14
s1
s2 <- starwars %>% slice_sample(n = 2)  #2x14
s2
```

```
# A tibble: 2 × 14
  name      height  mass hair_color skin_color eye_color bi
  <chr>      <int> <dbl> <chr>      <chr>      <chr>
1 Han Solo     180    80 brown      fair       brown
2 Obi-Wan …    182    77 auburn, w… fair       blue-gray
# ℹ 5 more variables: homeworld <chr>, species <chr>, films
#   vehicles <list>, starships <list>

# A tibble: 2 × 14
  name      height  mass hair_color skin_color eye_color bi
  <chr>      <int> <dbl> <chr>      <chr>      <chr>
1 Jabba De…    175  1358 <NA>       green-tan… orange
2 Yoda          66    17 white      green      brown
# ℹ 5 more variables: homeworld <chr>, species <chr>, films
#   vehicles <list>, starships <list>
```

Binding Rows

```
s1 <- starwars %>% slice_sample(n = 2)  #2x14
s1
s2 <- starwars %>% slice_sample(n = 2)  #2x14
s2
s3 <- starwars %>% slice_sample(n = 2)  #2x14
```

```
# A tibble: 2 × 14
  name      height  mass hair_color skin_color eye_color bi
  <chr>      <int> <dbl> <chr>      <chr>      <chr>
1 Shaak Ti     178    57 none       red, blue… black
2 Kit Fisto    196    87 none       green      black
# ℹ 5 more variables: homeworld <chr>, species <chr>, films
#   vehicles <list>, starships <list>

# A tibble: 2 × 14
  name      height  mass hair_color skin_color eye_color bi
  <chr>      <int> <dbl> <chr>      <chr>      <chr>
1 Palpatine    170  75   grey       pale       yellow
2 Luminara…    170  56.2 black      yellow     blue
# ℹ 5 more variables: homeworld <chr>, species <chr>, films
#   vehicles <list>, starships <list>
```

Binding Rows

```
s1 <- starwars %>% slice_sample(n = 2)  #2x14
s1
s2 <- starwars %>% slice_sample(n = 2)  #2x14
s2
s3 <- starwars %>% slice_sample(n = 2)  #2x14
s3
```

```
# A tibble: 2 × 14
  name      height  mass hair_color skin_color eye_color bi
  <chr>      <int> <dbl> <chr>      <chr>      <chr>
1 Sebulba      112    40 none       grey, red  orange
2 Rugor Na…    206    NA none       green      orange
# ℹ 5 more variables: homeworld <chr>, species <chr>, films
#   vehicles <list>, starships <list>

# A tibble: 2 × 14
  name      height  mass hair_color skin_color eye_color bi
  <chr>      <int> <dbl> <chr>      <chr>      <chr>
1 Gregar T…    185    85 black      dark       brown
2 San Hill     191    NA none       grey       gold
# ℹ 5 more variables: homeworld <chr>, species <chr>, films
#   vehicles <list>, starships <list>

# A tibble: 2 × 14
  name      height  mass hair_color skin_color eye_color bi
  <chr>      <int> <dbl> <chr>      <chr>      <chr>
1 Yoda          66    17 white      green      brown
2 Mon Moth…    150    NA auburn     fair       blue
# ℹ 5 more variables: homeworld <chr>, species <chr>, films
#   vehicles <list>, starships <list>
```

Binding Rows

```r
s1 <- starwars %>% slice_sample(n = 2)  #2x14
s1
s2 <- starwars %>% slice_sample(n = 2)  #2x14
s2
s3 <- starwars %>% slice_sample(n = 2)  #2x14
s3
star_wars_combined <- bind_rows(s1, s2, s3)  #6x14
```

```
# A tibble: 2 × 14
  name      height  mass hair_color skin_color eye_color bi
  <chr>      <int> <dbl> <chr>      <chr>      <chr>
1 Nien Nunb    160    68 none       grey       black
2 Jabba De…    175  1358 <NA>       green-tan… orange
# ℹ 5 more variables: homeworld <chr>, species <chr>, films
#   vehicles <list>, starships <list>

# A tibble: 2 × 14
  name      height  mass hair_color skin_color eye_color bi
  <chr>      <int> <dbl> <chr>      <chr>      <chr>
1 Kit Fisto    196    87 none       green      black
2 Lama Su      229    88 none       grey       black
# ℹ 5 more variables: homeworld <chr>, species <chr>, films
#   vehicles <list>, starships <list>

# A tibble: 2 × 14
  name      height  mass hair_color skin_color eye_color bi
  <chr>      <int> <dbl> <chr>      <chr>      <chr>
1 Boba Fett    183  78.2 black      fair       brown
2 Gregar T…    185  85   black      dark       brown
# ℹ 5 more variables: homeworld <chr>, species <chr>, films
#   vehicles <list>, starships <list>
```

Binding Rows

```r
s1 <- starwars %>% slice_sample(n = 2)  #2x14
s1
s2 <- starwars %>% slice_sample(n = 2)  #2x14
s2
s3 <- starwars %>% slice_sample(n = 2)  #2x14
s3
star_wars_combined <- bind_rows(s1, s2, s3)  #6x14
star_wars_combined
```

```
# A tibble: 2 × 14
  name      height  mass hair_color skin_color eye_color bi
  <chr>      <int> <dbl> <chr>      <chr>      <chr>
1 Lando Ca…    177    79 black      dark       brown
2 Quarsh P…    183    NA black      dark       brown
# ℹ 5 more variables: homeworld <chr>, species <chr>, films
#   vehicles <list>, starships <list>

# A tibble: 2 × 14
  name      height  mass hair_color skin_color eye_color bi
  <chr>      <int> <dbl> <chr>      <chr>      <chr>
1 Ben Quad…    163    65 none       grey, gre… orange
2 Beru Whi…    165    75 brown      light      blue
# ℹ 5 more variables: homeworld <chr>, species <chr>, films
#   vehicles <list>, starships <list>

# A tibble: 2 × 14
  name      height  mass hair_color skin_color eye_color bi
  <chr>      <int> <dbl> <chr>      <chr>      <chr>
1 Taun We      213    NA none       grey       black
2 Lando Ca…    177    79 black      dark       brown
# ℹ 5 more variables: homeworld <chr>, species <chr>, films
#   vehicles <list>, starships <list>

# A tibble: 6 × 14
  name      height  mass hair_color skin_color eye_color bi
  <chr>      <int> <dbl> <chr>      <chr>      <chr>
1 Lando Ca…    177    79 black      dark       brown
2 Quarsh P…    183    NA black      dark       brown
3 Ben Quad…    163    65 none       grey, gre… orange
4 Beru Whi…    165    75 brown      light      blue
```

Joining Data

```
fnames <- list.files(path = "data_headers", full.names
```

Joining Data

```
fnames <- list.files(path = "data_headers", full.names
header <- read_delim(fnames, col_names = c("field", "va
```

Joining Data

```
fnames <- list.files(path = "data_headers", full.names
header <- read_delim(fnames, col_names = c("field", "va
trials <- read_tsv(fnames, col_names = c("trial_num", '
```

Joining Data

```
fnames <- list.files(path = "data_headers", full.names
header <- read_delim(fnames, col_names = c("field", "va
trials <- read_tsv(fnames, col_names = c("trial_num", "
header
```

```
# A tibble: 12 × 8
   filename                Participant Age   Sex   Order Fi
   <chr>                   <chr>       <chr> <chr> <chr> <c
 1 data_headers/6191_1.txt 6191        25    male  1     4
 2 data_headers/6191_2.txt 6191        25    male  1     4
 3 data_headers/6191_3.txt 6191        25    male  1     4
 4 data_headers/6191_4.txt 6191        25    male  1     4
 5 data_headers/6191_5.txt 6191        25    male  1     4
 6 data_headers/6191_6.txt 6191        25    male  1     4
 7 data_headers/6192_1.txt 6192        21    male  2     4
 8 data_headers/6192_2.txt 6192        21    male  2     4
 9 data_headers/6192_3.txt 6192        21    male  2     4
10 data_headers/6192_4.txt 6192        21    male  2     4
11 data_headers/6192_5.txt 6192        21    male  2     4
12 data_headers/6192_6.txt 6192        21    male  2     4
```

Joining Data

```
fnames <- list.files(path = "data_headers", full.names
header <- read_delim(fnames, col_names = c("field", "va
trials <- read_tsv(fnames, col_names = c("trial_num", "
header
trials
```

```
# A tibble: 12 × 8
   filename                Participant Age   Sex   Order Fi
   <chr>                   <chr>       <chr> <chr> <chr> <c
 1 data_headers/6191_1.txt 6191        25    male  1     4
 2 data_headers/6191_2.txt 6191        25    male  1     4
 3 data_headers/6191_3.txt 6191        25    male  1     4
 4 data_headers/6191_4.txt 6191        25    male  1     4
 5 data_headers/6191_5.txt 6191        25    male  1     4
 6 data_headers/6191_6.txt 6191        25    male  1     4
 7 data_headers/6192_1.txt 6192        21    male  2     4
 8 data_headers/6192_2.txt 6192        21    male  2     4
 9 data_headers/6192_3.txt 6192        21    male  2     4
10 data_headers/6192_4.txt 6192        21    male  2     4
11 data_headers/6192_5.txt 6192        21    male  2     4
12 data_headers/6192_6.txt 6192        21    male  2     4

# A tibble: 240 × 5
   filename                trial_num speed_actual speed_res
   <chr>                       <dbl> <chr>        <chr>
 1 data_headers/6191_1.txt         1 fas          slower
 2 data_headers/6191_1.txt         2 fas          faster
 3 data_headers/6191_1.txt         3 fas          faster
 4 data_headers/6191_1.txt         4 fas          slower
 5 data_headers/6191_1.txt         5 fas          faster
 6 data_headers/6191_1.txt         6 slo          slower
 7 data_headers/6191_1.txt         7 fas          faster
 8 data_headers/6191_1.txt         8 slo          slower
 9 data_headers/6191_1.txt         9 slo          slower
10 data_headers/6191_1.txt        10 slo          faster
# ℹ 230 more rows
```

Joining Data

```
fnames <- list.files(path = "data_headers", full.names
header <- read_delim(fnames, col_names = c("field", "va
trials <- read_tsv(fnames, col_names = c("trial_num", "
header
trials
left_join(header, trials, by = "filename")
```

```
# A tibble: 12 × 8
   filename                Participant Age   Sex   Order Fi
   <chr>                   <chr>       <chr> <chr> <chr> <c
 1 data_headers/6191_1.txt 6191        25    male  1     4
 2 data_headers/6191_2.txt 6191        25    male  1     4
 3 data_headers/6191_3.txt 6191        25    male  1     4
 4 data_headers/6191_4.txt 6191        25    male  1     4
 5 data_headers/6191_5.txt 6191        25    male  1     4
 6 data_headers/6191_6.txt 6191        25    male  1     4
 7 data_headers/6192_1.txt 6192        21    male  2     4
 8 data_headers/6192_2.txt 6192        21    male  2     4
 9 data_headers/6192_3.txt 6192        21    male  2     4
10 data_headers/6192_4.txt 6192        21    male  2     4
11 data_headers/6192_5.txt 6192        21    male  2     4
12 data_headers/6192_6.txt 6192        21    male  2     4

# A tibble: 240 × 5
   filename                trial_num speed_actual speed_res
   <chr>                       <dbl> <chr>        <chr>
 1 data_headers/6191_1.txt         1 fas          slower
 2 data_headers/6191_1.txt         2 fas          faster
 3 data_headers/6191_1.txt         3 fas          faster
 4 data_headers/6191_1.txt         4 fas          slower
 5 data_headers/6191_1.txt         5 fas          faster
 6 data_headers/6191_1.txt         6 slo          slower
 7 data_headers/6191_1.txt         7 fas          faster
 8 data_headers/6191_1.txt         8 slo          slower
 9 data_headers/6191_1.txt         9 slo          slower
10 data_headers/6191_1.txt        10 slo          faster
# ℹ 230 more rows
```

Joining Data

```
fnames <- list.files(path = "data_headers", full.names
header <- read_delim(fnames, col_names = c("field", "va
trials <- read_tsv(fnames, col_names = c("trial_num", "
header
trials
left_join(header, trials, by = "filename")
left_join(slice_sample(header, n = 5), trials, by = "fi
```

```
# A tibble: 12 × 8
   filename                Participant Age   Sex   Order Fi
   <chr>                   <chr>       <chr> <chr> <chr> <c
 1 data_headers/6191_1.txt 6191        25    male  1     4
 2 data_headers/6191_2.txt 6191        25    male  1     4
 3 data_headers/6191_3.txt 6191        25    male  1     4
 4 data_headers/6191_4.txt 6191        25    male  1     4
 5 data_headers/6191_5.txt 6191        25    male  1     4
 6 data_headers/6191_6.txt 6191        25    male  1     4
 7 data_headers/6192_1.txt 6192        21    male  2     4
 8 data_headers/6192_2.txt 6192        21    male  2     4
 9 data_headers/6192_3.txt 6192        21    male  2     4
10 data_headers/6192_4.txt 6192        21    male  2     4
11 data_headers/6192_5.txt 6192        21    male  2     4
12 data_headers/6192_6.txt 6192        21    male  2     4

# A tibble: 240 × 5
   filename                trial_num speed_actual speed_res
   <chr>                       <dbl> <chr>        <chr>
 1 data_headers/6191_1.txt         1 fas          slower
 2 data_headers/6191_1.txt         2 fas          faster
 3 data_headers/6191_1.txt         3 fas          faster
 4 data_headers/6191_1.txt         4 fas          slower
 5 data_headers/6191_1.txt         5 fas          faster
 6 data_headers/6191_1.txt         6 slo          slower
 7 data_headers/6191_1.txt         7 fas          faster
 8 data_headers/6191_1.txt         8 slo          slower
 9 data_headers/6191_1.txt         9 slo          slower
10 data_headers/6191_1.txt        10 slo          faster
# ℹ 230 more rows
```

Joining Data

```
fnames <- list.files(path = "data_headers", full.names
header <- read_delim(fnames, col_names = c("field", "va
trials <- read_tsv(fnames, col_names = c("trial_num", "
header
trials
left_join(header, trials, by = "filename")
left_join(slice_sample(header, n = 5), trials, by = "fi
left_join(slice_sample(trials, n = 5), header, by = "fi
```

```
# A tibble: 12 × 8
   filename                Participant Age   Sex   Order Fi
   <chr>                   <chr>       <chr> <chr> <chr> <c
 1 data_headers/6191_1.txt 6191        25    male  1     4
 2 data_headers/6191_2.txt 6191        25    male  1     4
 3 data_headers/6191_3.txt 6191        25    male  1     4
 4 data_headers/6191_4.txt 6191        25    male  1     4
 5 data_headers/6191_5.txt 6191        25    male  1     4
 6 data_headers/6191_6.txt 6191        25    male  1     4
 7 data_headers/6192_1.txt 6192        21    male  2     4
 8 data_headers/6192_2.txt 6192        21    male  2     4
 9 data_headers/6192_3.txt 6192        21    male  2     4
10 data_headers/6192_4.txt 6192        21    male  2     4
11 data_headers/6192_5.txt 6192        21    male  2     4
12 data_headers/6192_6.txt 6192        21    male  2     4

# A tibble: 240 × 5
   filename                trial_num speed_actual speed_res
   <chr>                       <dbl> <chr>        <chr>
 1 data_headers/6191_1.txt         1 fas          slower
 2 data_headers/6191_1.txt         2 fas          faster
 3 data_headers/6191_1.txt         3 fas          faster
 4 data_headers/6191_1.txt         4 fas          slower
 5 data_headers/6191_1.txt         5 fas          faster
 6 data_headers/6191_1.txt         6 slo          slower
 7 data_headers/6191_1.txt         7 fas          faster
 8 data_headers/6191_1.txt         8 slo          slower
 9 data_headers/6191_1.txt         9 slo          slower
10 data_headers/6191_1.txt        10 slo          faster
# ℹ 230 more rows
```